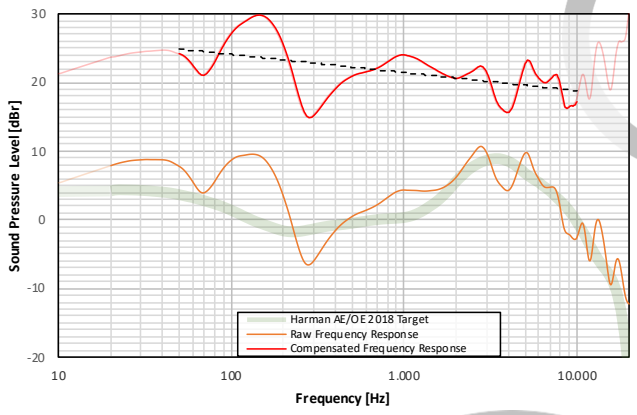
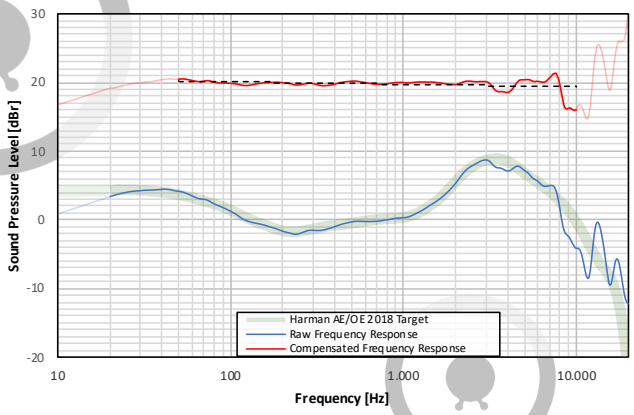
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# EQ setting for Sennheiser HD820

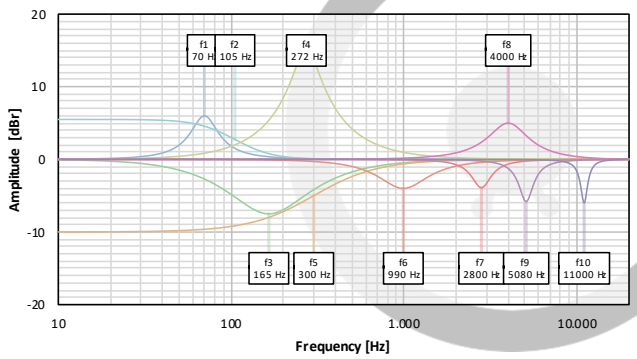
SPL Frequency Response without EQ



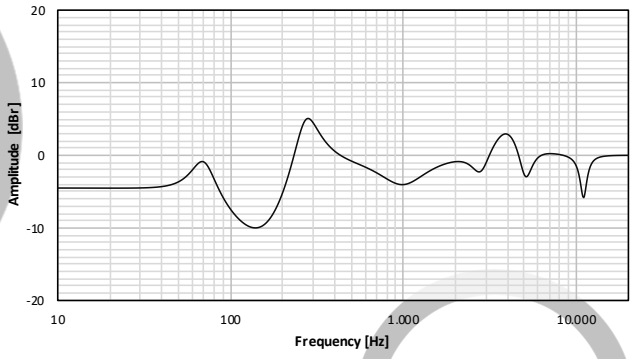

SPL Frequency Response with EQ



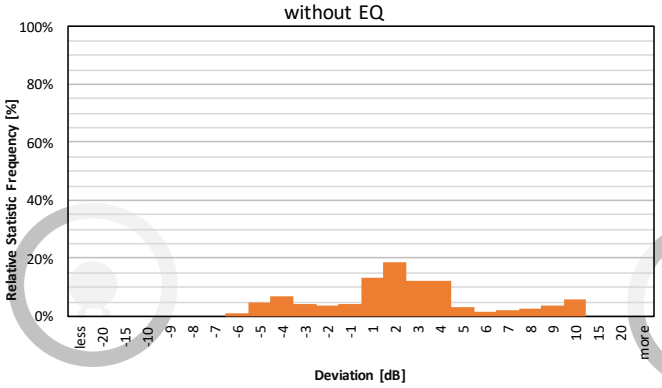

EQ Curve Individual Filters



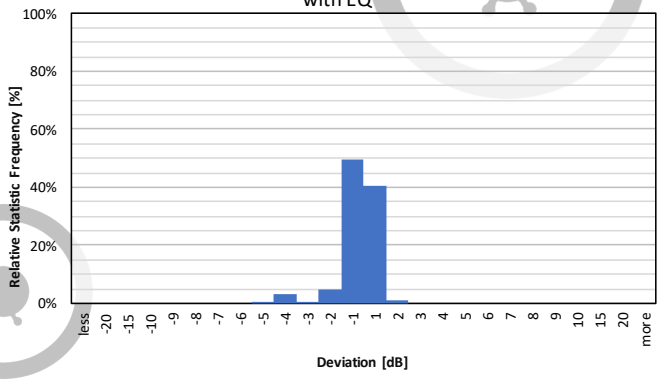

EQ Curve total




Error Curve Histogram without EQ




Error Curve Histogram with EQ




| Filter Settings | | | | | |
|---|---|---|---|---|---|
| | Filter Type | Frequency | Gain | Q-Factor | BW |
| Band 1 | PEAK | 70 Hz | 6,0 dB | 2,1 | 0,68 |
| Band 2 | LOW_SHELF | 105 Hz | 5,5 dB | 0,71 | 1,89 |
| Band 3 | PEAK | 165 Hz | -7,5 dB | 0,7 | 1,92 |
| Band 4 | PEAK | 272 Hz | 15,3 dB | 1,4 | 1,01 |
| Band 5 | LOW_SHELF | 300 Hz | -10,0 dB | 0,55 | 2,35 |
| Band 6 | PEAK | 990 Hz | -4,0 dB | 1,2 | 1,17 |
| Band 7 | PEAK | 2800 Hz | -3,9 dB | 3,0 | 0,48 |
| Band 8 | PEAK | 4000 Hz | 5,0 dB | 1,4 | 1,01 |
| Band 9 | PEAK | 5080 Hz | -5,8 dB | 3,8 | 0,38 |
| Band 10 | PEAK | 11000 Hz | -6,0 dB | 6,0 | 0,24 |

| Preamp gain: |
|---|
| -5,1 dB |

| Deviation from Target | |
|---|---|
| Before EQ | After EQ |
| 3,23 dB | 0,41 dB |

| Preference Rating* | |
|---|---|
| Before EQ | After EQ |
| 51/100 | 100/100 |

Adjust gain of band 2 to preference (bass)

*preference rating prediction based on:
[1] S. Olive et al: "A Statistical Model That Predicts Listeners' Preference Ratings of In-Ear Headphones: Part 1" (2017)
[2] S. Olive et al: "A Statistical Model That Predicts Listeners' Preference Ratings of In-Ear Headphones: Part 2" (2017)
[3] S. Olive et al: "A Statistical Model That Predicts Listeners' Preference Ratings of Around-Ear and On-Ear Headphones" (2018)
*The normalized preference ratings are used, where zero deviation from target equals a preference rating of 100*

measured on GRAS45BC-10 KEMAR
GRAS KB5000/5001 anthropometric pinnae
APx515 or APx526 Signal Generator/Analyzer
APx1701 amplifier

Calculator App v2.12
by oratory1990
05.07.24